End-to-End Experimental Validation Workflow: AST-Route Testbed (75 Queries, 6 Engines)

PHASE 1: Full-Scale TPC-H Data Provisioning (SF = 1 GB)

- Base Dataset: 8 Relational Tables (8.66M rows; lineitem = 6.00M rows)
- Local Columnar: ZSTD Parquet + DuckDB Native + Apache Arrow Buffers
- Local Row-Store Baseline: SQLite 3 Full B-Tree with PK/FK/Date Indexes
- Cloud Serverless: Google Cloud Storage -> BigQuery Unopt vs. Clustered

PHASE 2: 75-Query Factorial Workload Design (25 Seeds x 3 Horizons)

- 5 Structural Tiers (5 Canonical Seeds/Tier = 25 Disjoint Seed Groups):
  L1 (Scan/Filter), L2 (Multi-Join), L3 (Staged CTE), L4 (Window), L5 (UNNEST)
- 3 Selectivity Perturbations per Seed (Total N = 75 Queries):
  V1: High Selectivity (1-5%) | V2: Medium (15-30%) | V3: Full-Horizon (65-95%)

PHASE 3: Static AST Feature Profiling, CSCI Construct Validation & Hardware Telemetry Instrumentation

- Parses all 75 SQL queries via SQLGlot into 9D vectors $x_q = [D, N, J, S, W, U, C, CSCI, \sigma(q)]^T$ (Mean CSCI = 29.66; Range: 15.3 in L1 to 37.1 in L4)
- CSCI Construct Validation: Spearman $\rho = 0.4935$ ($p = 6.84e\text{-}6$) vs. GPU Speedup; 1,000-Trial Monte Carlo Weight Perturbation Stability $\rho = 0.8189$
- High-Frequency Sensor Polling (250 Hz psutil + NVIDIA NVML + cudaEvent): Captures Latency (ms), CPU/GPU (%), RAM/VRAM (GB), Energy (J), Perf/Watt

Paradigm 1: Row-Store

- Engine: SQLite 3 (ANSI C)
- Storage: Full-Scale B-Tree
- Tuple-at-a-Time Volcano Model
- Mean Latency: 8417.33 ms
- 95% CI: [6412.2, 10819.7] ms
- Mean Energy: 126.26 J (4.05 MB/J)

Paradigm 2: Serverless Cloud

- Engine: Google BigQuery
- Capacitor Columnar Storage
- Cache Strictly Disabled
- Clustered Mean: 745.39 ms
- 95% CI: [515.6, 1026.5] ms
- V1 I/O Pruning: 78.4% ($1.47/1M)

Paradigm 3: Vectorized CPU

- 1) DuckDB (C++): 183.74 ms
  95% CI: [110.7, 274.9] ms (6.40 J)
- 2) Polars (Rust): 82.47 ms
  95% CI: [67.9, 98.7] ms (3.17 J)
- 3) DataFusion (Rust): 96.30 ms
  95% CI: [78.3, 114.9] ms (3.39 J)

Paradigm 4: SIMT CUDA GPU

- Engine: PyTorch CUDA Tensor
- GPU: NVIDIA RTX 2050 (4GB)
- Cold Mean: 56.96 ms (0.65 J)
  95% CI: [46.23, 68.44] ms
- Warm Mean: 22.26 ms (0.25 J)
  95% CI: [16.78, 28.68] ms

PHASE 5: Statistical Inference & 25-Seed LOSO-CV Routing

- Exact Equivalence EX = 1 (100%) | Friedman $\chi^2_F(5) = 321.50$ ($p = 2.38e\text{-}67$)
- 10,000-Bootstrap Seed-Clustered 95% CIs & Cliff's Delta Effect Sizes
- 5-Policy Seed-Grouped LOSO-CV: AST-Route Achieves 54.92 ms (34.98% Regret)

PHASE 6: Cryptographic SHA-256 Audit & Reproducibility

- Automated Booktabs Tables (Tables 1-8) & Unified 300 DPI Vector Figures
- Zero-Discrepancy Verification Suite & SHA-256 Telemetry Audit Certificate
- Open-Source Benchmark & Router Repository: github.com/preecha-msu